

Roll No. _____

22/2104
B.C.A. Examination, 2022
Fifth Semester
First Paper
(Introduction to DBMS)

*Time : 1:30 Hours]**[Maximum Marks : 75*

Note : Attempt any **five** questions. **All** questions carry equal marks. The answer to given questions should not exceed 250 words each. 15×5=75

1. What is functional dependency? Discuss its types, explain the role of functional dependency in the process of Normalization.
2. What is E-R model? How E-R model different from other data models? What are the main advantages of the E-R model?

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3. What **are** the various operators used in relational Algebra? Discuss each with an example.
4. What do you understand by relational model of database system? Elaborate the major characteristics of relational database management system.
5. (a) Explain the responsibilities and functions of database Administrator.
(b) Differentiate between physical and logical data independence.
6. Elaborate the three level architecture of database system.
7. What are the problems arising out of concurrent data access? What are the various methods of handling concurrency?
8. Write note on the following :
 - (a) Normalization
 - (b) Hashing
 - (c) Inheritance

9. Consider the following relations :

Hotel {hotel No., name, address}

Room {room No., hotel No., type, price}

Booking {hotel No., guest No., date From,
date to, room No.}

Guest {guest No., name, address}

Write the SQL statements for the following:

- (a) List the names and addresses of all guest in delhi, alphabetically ordered by name.
- (b) List all family rooms with a price below Rs. 400 per night, in ascending order of price.
- (c) How many hotels are there?